Aaryaa Moharir
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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — *May 2028*
Master's of Artificial Innovation Engineering – Information Security — *GPA: 4.0/4.0*
Coursework: Trustworthy AI, AI App Information Security, Systems Tool Chain in AI

University of Texas at Dallas, Richardson, TX — *May 2026*
Bachelor of Computer Science — *GPA: 4.0/4.0*
Coursework: Cyber Defense Lab, Database Systems, Adv. Algorithms, Data Structures & Algorithms, Operating Systems

SKILLS

Languages: Java, Python, C#, C++ | **Security:** MITRE ATT&CK, Penetration Testing, Malware Analysis, Data Security, CTFs, OWASP Top 10
Tools: Wireshark, Metasploit, MongoDB, SQL, Linux, AWS | **Development:** React, JavaScript, REST APIs, Database Design

EXPERIENCE

Incoming Security Engineering Intern – Amazon — June 2026 – August 2026

- Designed a security framework for Agentic AI workflows across SDO and AWS, modeling and mitigating 32 security threats.
- Made testing playbooks, threat model templates, and conducted security reviews using the framework.
- Engineered agent scripts per threat for automated vulnerability detection.
- Worked with scientists to implement an internal benchmarking framework to systematically evaluate agentic AI skills, defining evaluation datasets, performance metrics, and automated test pipelines to measure precision, recall, accuracy, and f1 score.

Cybersecurity Research Intern – DNIF Hypercloud — June 2025 – August 2025

- Researched and profiled **Lumma Stealer malware**, mapping its multi-stage attack lifecycle to **MITRE ATT&CK** and documenting key IOCs across host and network activity.
- Built a **Detection Use Case Matrix** and **Threat Hunting Playbook**, authoring pseudo-Sigma detection logic across event IDs and log sources to flag process hollowing, LSASS access, and HTTP-based exfiltration.
- Delivered a **Detection Coverage Map** identifying visibility gaps across 8 use cases and presenting remediation recommendations.

Undergraduate Researcher in the Intelligent Systems Lab – UT Dallas — May 2024 – December 2025

- Implemented **gradient-based, semantic-preserving input adaptation techniques** to refine images for uncertain classifications, improving model performance through contrast adjustment, rotation, and validation.
- Contributed to research on GPU bugs in tile programs, analyzing **900+ samples** to map root causes to symptoms, and developed a **structured framework** for automated bug detection and repair, including the preliminary tool **Tile-CBDetect**.
- Applied a **methodical research approach** to unfamiliar problem domains, building structured understanding before designing solutions, reinforcing expertise in AI-driven security and automated analysis.

Undergraduate Researcher in the System Security Lab – UT Dallas — August 2024 – May 2025

- Contributed to the development of attack pathway detection techniques by analyzing network traffic such as file access, IP connections, and process activity to identify malicious behavior patterns and implemented 6 tracking strategies using C#.
- Built a database system to aggregate and query event counts using MongoDB, SQL, and C#, reducing data query time by **30%** for large-scale network analysis.
- Integrated and adapted the MAGIC anomaly detection model for threat identification, customizing the pipeline for compatibility with the lab's proprietary dataset.

PROJECTS

Capture The Flag Participant – UT Dallas — January 2025 – May 2025

- Completed 70+ CTF challenges spanning exploit development (stack/heap/buffer overflows, shellcode) and binary reversing, building hands-on offensive/defensive tradecraft and scoring in the top 10%.

ECHO 911 | Winner (1st place), Breaking Barriers Hackathon (AWS, AT&T, Deloitte) — September 2025- September 2025

- Built a real-time dispatcher dashboard using React (frontend) and Python (backend) to display 911 transcripts, outline policies to follow on-scene, and support dispatcher decision-making.
- Leveraged AWS services (Connect, Lambda, Bedrock, Strands, S3, Polly) to transcribe calls, extract key details, and evaluate incident severity in our **agentic application**.
- Integrated OpenStreetMap API, Open-Meteo API, Amazon Bedrock, and Claude Sonnet to automate first responder mapping.

Detect and Defend – Team Lead — December 2023 – May 2024

- Led and mentored a 5-person team developing a Python-based machine learning model to attack Federated Learning systems by reconstructing original training datasets through gradient reversal techniques.
- Utilized LeNet-5 architecture with multiple datasets including MNIST to train and evaluate a General Regression Neural Network model.
- Demonstrated security vulnerabilities in distributed learning systems and proposed defensive countermeasures.

Ransomware Attack Simulation — February 2022 – May 2022

- Leveraged Metasploit, a widely used penetration testing tool, for generating a reverse shell to access the victim device on a virtual machine.
- Built and deployed a Python-based keylogger that bypassed the Windows firewall, a HTML-based phishing website, and Java based ransomware to effectively crash the victim's computer and presented to 30+ people to inform them about cybersecurity.

ACCOMPLISHMENTS & CERTIFICATES

- 1st place out of 36 teams at the Breaking Barriers Hackathon hosted by AWS, AT&T, and Deloitte — September 2025
- GIAC Foundational Cybersecurity Technologies (GFACT) Certified — July 2023
- Published "Characterizing Real-World Bugs in Tile Programs for Automated Bug Detection" — May 2025